

Phase 4 – Test Execution and Defect Management:

6.1 Test Execution Summary Table:

Module	Run #	Date	Total TCs	Passed	Failed	Blocked	Not Run	Pass Rate
Account Management	Run 1	06/07/2026	46	43	3	0	0	93,48%
Authentication	Run 1	06/07/2026	32	29	2	0	1	90,63%
Validation Library	Run 1	06/07/2026	48	43	5	0	0	89,58%
Transaction Engine	Run 1	06/07/2026	60	57	3	0	0	95,00%
Reporting Engine	Run 1	06/07/2026	10	10	0	0	0	100,00%
Interest Calculator	Run 1	06/07/2026	62	47	15	0	0	75,81%
Loan Processing	Run 1	06/07/2026	23	20	3	0	0	86,96%

OVERALL TEST EXECUTION SUMMARY							
Run #	Date	Total TCs	Passed	Failed	Blocked	Not Run	Pass Rate
Run 1 Baseline	06/07/2026	281	249	31	0	1	88,61%
Run 2 After Fix 1	07/07/2026	31	31	0	0	0	100,00%
Run 3 Regression	08/07/2026	115	115	0	0	0	100,00%

6.2 Defect Management:

Please refer to the DefectLog.xlsx for the complete log of all 17 bugs found in BankCoreSolution's program

6.3 Severity Vs. Priority:

Defect ID: BUG-009

Title: System allows transfers to me made to closed or dormant accounts

Severity: Critical

Priority: P1

Explanation: BUG-009 has both a critical severity rating and a P1 priority rating because this bug depicts a complete failure of banking logic that can lead to financial loss by allowing funds to be transferred to an account that potentially does not exist anymore. This bug has a critical severity rating because allowing funds to be transferred to a closed or dormant account can cause huge losses and errors. It also has a P1 priority rating because this bug would need to be fixed as soon as possible, as this could lead to a decline customer trust, and loss.

Defect ID: BUG-010

Title: DTI is incorrectly calculated by not including monthly installment in the calculation

Severity: High

Priority: P1

Explanation: In the case of BUG-010, it has a High Severity Rating, and a P1 priority rating because the program still functions correctly, but performs an incorrect calculation. In the case of a banking application like the BankCoreSolutions app, having the system incorrectly allow a loan to be submitted, and potentially approved because the Debt-To-Income ratio was not correctly calculated poses a huge risk because this allows individuals to be approved for a loan which they cannot actually afford to repay, which is why this bug has a High severity rating and a P1 priority rating.

Defect ID: BUG-013

Title: The application does not allow a locked account to log back in, even after the lockoutExpiry time has expired

Severity: Medium

Priority: P2

Explanation: BUG-013 has a Medium severity rating, but a higher P2 priority rating because BUG-013 is isolated to individual users, and does not cause data loss. While having the risk of being completely locked out of the program is not high, but still exists, which is why it has a P2 priority rating to be fixed. The sooner this bug gets fixed, the lesser chance there is for complete system lockout.

6.4 Regression Testing:

After the bugs had been fixed with simulated fixes for all of the modules that contained bugs, a regression cycle was run again.

- Test Cases Included: All test cases for all bug-infected modules have been rerun to ensure that no regression occurred.
- Rationale for including test cases: After the core code of the program has been altered to rectify the bugs, several critical financial formulas were altered to be fixed. It was of utmost critical importance to re-run all of the tests on all of the modules to ensure that no regression had taken place, and more importantly, to ensure that the application now performs each task as it should, and to ensure that all the bugs have been resolved.
- Regression status: No regression was introduced after the simulated fixes were implemented. All of the tests that passed previously remained passing, and all of the tests that failed also passed.

6.5 Non-Functional Test Execution:

6.5.1 Performance Test Results

Test Case ID	Title	Threshold	Measured time	Pass/Fail	Recommendation
TC-BEMS-PERF-001	How long it takes for the program to run	The program should launch quickly, within ~5 seconds within clicking the login button	~2 seconds to launch the program	Pass	
TC-BEMS-PERF-002	The time it takes to log into the system	The program should login within 0.5 seconds of clicking the login button	0.5 seconds to log into the program	Pass	
TC-BEMS-PERF-003	How long it takes to generate a account transaction history with 200+ transactions	15 to 20 seconds for a large report	Report generated in 12 seconds	Pass	
TC-BEMS-PERF-004	How long transactions take when multiple tellers are performing transactions concurrently	Regular transactions should take around 2 seconds to process	Transactions process within 1.5 seconds with multiple transactions being performed at the same time	Pass	
TC-BEMS-PERF-005	Testing how long the system takes to run a system audit log of 1 year	System should take around 10 to 15 minutes to generate the detailed system audit log	The system took 14 minutes to generate the complete systemlog	Pass	
TC-BEMS-PERF-006	Testing to see how well the program performs after 24 hours of continuous operations being performed on it	The program should see a very slight degradation in performance, and should take around 1 to 2 seconds to process transactions	Program took 1 second to process transactions	Pass	

6.5.2 Security Test results

Test Case ID	Test performed	Expected behavior	Actual behavior	Risk rating (if failed)
TC-BEMS-SEC-001	Logging in with invalid credentials	The program should not allow the user to login	The program does not allow the user to log in	
TC-BEMS-SEC-002	Attempting to access admin features as a standard user	The program should not allow the standard user to access administrator features	The program does not allow the standard user to access admin features	
TC-BEMS-SEC-003	Session tokens expire upon logout	The program should expire the user session token upon logout	Then token is set to expire	
TC-BEMS-SEC-004	SQL string injection prevention	The program should reject all SQL injection strings	The program rejects SQL injection strings	
TC-BEMS-SEC-005	Strong password requirements	Program should have strong password requirements, and not allow past passwords to be used	The program has strong password requirements	
TC-BEMS-SEC-006	Password hashing	Passwords should be hashed, and not displayed as plain text	All passwords are hashed	

6.5.3 Robustness Test Results

Test Case ID	Input used	Expected handling	Actual result	Pass/Fail
TC-BEMS-ROBUST-001	Numbers in a string only field	The program should display an error message to the user stating that only alphabetical characters are allowed	The program successfully rejects the input, and displays a message stating Alphabetic characters only	Pass
TC-BEMS-ROBUST-002	Entering alphabetical characters into a numerical only field	The system should reject the input, and display an error message static invalid input	The program displays an error message stating that only numerical fields are allowed	Pass
TC-BEMS-ROBUST-003	Attempting to transfer the maximum possible amount to transfer	The program should not crash, and instead should display an error	The program does not crash when the invalid amount is entered	Pass

	without the program crashing	message to the user stating the max amount they are allowed to enter		
TC-BEMS-ROBUST-004	Entering a ID number that does not meet the correct length or format	The program should handle this input error without the system crashing	The system displays an error message stating that the ID number entered is not valid.	Pass
TC-BEMS-ROBUST-005	Entering a password below the minimum allowed length	The system should reject the password, and display an error message stating that the password does not meet length requirements	The program displays an error message stating that the password does not meet the length requirements	Pass